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Master's Thesis

Interventions in Asymmetric Belief Systems

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in the

Computational Science Lab
Informatics Institute

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Declaration of Authorship

I, Henry Zwart, declare that this thesis, entitled 'Interventions in Asymmetric Belief Systems' and the work presented in it are my own. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty of Science
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Master of Science in Computational Science

Interventions in Asymmetric Belief Systems

by Henry Zwart

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Thank the people that have helped: supervisors, family, etc.

Use of AI

Statement that generative AI was not used in the writing of this thesis.

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Abbreviations

CSL **C**omputational **S**cience **L**ab

UvA **U**niversiteit **v**an **A**msterdam

Chapter 1

Terminology and notation

Define key terminology, outline notation used in following sections.

Chapter 2

Introduction

2.1 Plan

- Motivate the problem
- Contributions
- Research questions (thinking these are perhaps better left to later, in favour of contributions here)

2.2 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to: (i) support asymmetric causal effects between beliefs and attitudes, and (ii) model intervention dynamics.

Inferring belief systems:

- **RQ2.1:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?

- **RQ2.2:** How do (symmetric or asymmetric) belief systems relating to climate change vary between conservative and liberal individuals?

Intervention dynamics:

- **RQ3.1:** How do asymmetric and symmetric beliefs systems inferred from the climate attitudes dataset differ with regards to intervention strategy and effectiveness?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or between conservative and liberal individuals?

2.3 Proposed thesis structure

- **Terminology and notation**

- **Introduction:**
  - Motivate the problem,
  - Outline contributions (research questions)

- **Asymmetric belief system:**
  - Define and illustrate the model
  - Model simulation with Glauber dynamics
  - How do we model interventions?

- **Methods:**
  - Counterfactual intervention experiments — comparing against the no-intervention scenario, measuring differences in effects.
  - Parameter estimation:
    - Maximum likelihood estimation
      - Conditional on a specific binarisation
      - Marginalising over binarisation process
    - Regularisation

- **Existence and impacts of asymmetry in belief systems**
  - Results for:
    - **RF1:** Show asymmetric model fit
    - **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
    - **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).

- **Individual heterogeneity in belief systems and intervention dynamics**
  - Second results section
  - **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.

- **RQ3.2:**
  - Distribution of intervention effectiveness by individual.
  - How does intervention ranking vary across individuals?
  - Characterising how different initial states affect intervention success.
  - Looking at how other theory-driven features affect success, e.g.:

    - How receptive is the individual to the intervention?

      The effective baseline $(h_i + \sum_j J_{ji} s_j)$ determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

      We could look at how this changes with different intervention strengths (it follows a logistic curve).

    - How consistent is the individual's belief state, as measured by the total system energy?

    - How 'entrenched' is the target attitude?

      Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

- **General discussion**
  - Draw out the key findings from the above two results sections
  - Make clear the implications for our theory of belief system dynamics
  - Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
  - Representational limitations of our model:
    - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
    - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
    - More generally, effect sizes may vary depending on specific spin states.

- **Dataset**
  - Introduce the (insert name here) dataset, give context, survey details
  - Validation, cleaning, transformations
  - Question selection, indexes

  - Binarisation

- **Literature review/related work**

- **Conclusions and future work**

Chapter 3

Non-equilibrium belief systems

3.1 Belief system dynamics

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.

2. The state of a belief or attitude is Markovian with respect to the previous system configurations.

3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i}. The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \tag{3.1}$$

with conditional state probability $P\left(S_i^{\tau+1} \mid \boldsymbol{S}^1, ..., \boldsymbol{S}^{\tau}\right) = P\left(S_i^{\tau+1} \mid \boldsymbol{S}^{\tau}\right)$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j, the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\boldsymbol{S}^t$. We can therefore straightforwardly extend Equation 3.1 to describe the evolution of the entire belief system as a Markov process,

$$\{\boldsymbol{s}^t\}_{t=1}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_{S_1} \times \cdots \times \Omega_{S_n} \tag{3.2}$$

with the probability of each belief system configuration given by the conditional distribution $P\left(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t\right)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

3.2 Non-equilibrium belief system model

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that $P\left(\boldsymbol{S}^t = \boldsymbol{s} \mid \boldsymbol{S}^{t-1}\right) = P(\boldsymbol{S} = \boldsymbol{s})$ for all configurations $\boldsymbol{s}$ and observation times t. The probability of observing $\boldsymbol{s}$ is then time-invariant and given by the Boltzmann probability parameterised by the Hamiltonian $H(\boldsymbol{s})$ (Cardy, 1996; Christensen & Moloney, 2005):

$$P(\boldsymbol{S} = \boldsymbol{s}) = \frac{e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s})}}{\sum_{\boldsymbol{s}'} e^{-\frac{1}{k_b T} \cdot H(\boldsymbol{s}')}} \tag{3.3}$$

where k_b is the Boltzmann constant, and the temperature parameter $T \in \mathbb{R}^+$ controls stochasticity. Equation 3.3 converges to a uniform distribution over all configurations when $T \to +\infty$, and a uniform distribution over the restricted set of configurations for which $H(\boldsymbol{s}')$ is minimised when $T \to 0^+$.

For the purposes of this study, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\boldsymbol{S}^t = \boldsymbol{s} \mid \boldsymbol{S}^{t-1})$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\boldsymbol{S}^{t-1}$ (Chapter 3.1). It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time t, as conditional on $\boldsymbol{S}^{t-1}$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamilonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i.

Other spins influence S_i's state through alignment or opposition relations. For a spin S_j, we define an interaction effect J_{ji} (read *'influence of j on i'*). When J_{ij} if positive or negative, S_i is influenced to adopt the state S_j^{t-1} or $-S_j^{t-1}$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* effect. Positive self-influence effects reflect the inertia of S_i, i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. The inclusion of self-influence effects allows us to capture the timescales of different beliefs or attitudes independently.

The energy experienced by S_i for a particular spin state $s \in \{0, 1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i:

$$H_i(s \mid \boldsymbol{s}^{t-1}) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^{t-1} \cdot s \tag{3.4}$$

where $\boldsymbol{A} \in \{0,1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i. Dividing the common factor of s, we can equivalently write Equation 3.4 as

$$H_i(s \mid \boldsymbol{s}^{t-1}) = s \cdot h_i^{\text{eff}}(\boldsymbol{s}^{t-1}) \tag{3.5}$$

where $h_i^{\text{eff}}(\boldsymbol{s}^{t-1}) = h_i + \sum_j A_{ji} J_{ji} s_j^{t-1}$ is the effective baseline activation experienced by S_i at time t. We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P(S_i^t = s \mid \boldsymbol{S}^{t-1} = \boldsymbol{s}) &= \frac{e^{\frac{1}{T}H_i(s \mid \boldsymbol{s})}}{\sum_{s' \in \{0,1\}} e^{\frac{1}{T}H_i(s' \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T}H_i(s \mid \boldsymbol{s})}}{e^{\frac{1}{T}H_i(s \mid \boldsymbol{s})} + e^{-\frac{1}{T}H_i(-s \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T}s \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T}s \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T}s \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \end{aligned} \tag{3.6}$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^{\boldsymbol{s}} := P(S_i^t = +1 \mid \boldsymbol{S}^{t-1} = \boldsymbol{s}) &= \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \text{logistic}\big(2h_i^{\text{eff}}(\boldsymbol{s})/T\big) \end{aligned} \tag{3.7}$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P(\boldsymbol{S}^t = \boldsymbol{s}^t \mid \boldsymbol{S}^{t-1} = \boldsymbol{s}^{t-1}) &= \prod_{i=1}^{n} P(S_i^t = s_i \mid \boldsymbol{S}^{t-1} = \boldsymbol{s}^{t-1}) \\ &= \prod_{i=1}^{n} \left[\left(\frac{1 + s_i}{2} \right) p_i^{\boldsymbol{s}^{t-1}} + \left(\frac{1 - s_i}{2} \right) \left(1 - p_i^{\boldsymbol{s}^{t-1}} \right) \right] \end{aligned} \tag{3.8}$$

NOTE: The initial distribution must be specified, since the model only captures transition probabilities.

3.3 Symmetric and asymmetric belief systems

Let $\mathcal{M}$ be a belief system model defined as in the preceding section, comprising $N \in \mathbb{N}$ beliefs or attitudes, with adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, interaction effect matrix $\boldsymbol{J} \in \mathbb{R}^{N \times N}$, and baseline activations $\boldsymbol{h} \in \mathbb{R}^N$.

When each entry in $\boldsymbol{J}$ is independent of all others, we say that $\mathcal{M}$ is an **asymmetric belief system**. The term *asymmetric* here refers to the directed relations between a pair of nodes. In an asymmetric belief system, for any pair of distinct nodes $S_i \neq S_j$, it may be the case that an influence relation exists only in one direction, or that the directed relations differ in magnitude.

In the special case where we constrain $A_{ij} = A_{ji}$ and $J_{ij} = J_{ji}$ for all $i, j \in [1, N]$, we instead say that $\mathcal{M}$ is a **symmetric belief system**. In a symmetric belief system all interactions are directionally-equivalent. This can be considered a simple extension of the symmetric Ising model with (i) self-interaction terms and (ii) temporal dynamics, thus the symmetric belief system model tends toward an equilibrium steady state.

3.4 Simulation via Glauber dynamics

It is straightforward to draw samples from a belief system model using Glauber dynamics, given the conditional probability distribution defined in Equation 3.8.

Given an initial state $\boldsymbol{s}^0 \in \{-1, +1\}^N$, we sample a sequence of $T \in \mathbb{N}$ subsequent configurations:

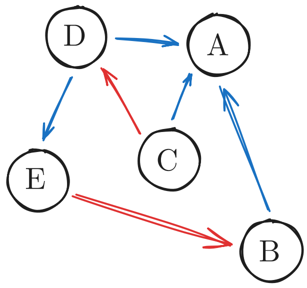
$$\{\boldsymbol{s}^t\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{s}^t \sim P(\boldsymbol{S}^t \mid \boldsymbol{S}^{t-1}) \tag{3.9}$$

Each belief or attitude has the opportunity to update during every time interval. This update routine is referred to as *synchronous* Glauber dynamics, contrasting *asynchronous* Glauber dynamics, in which only one spin can update during a given interval.

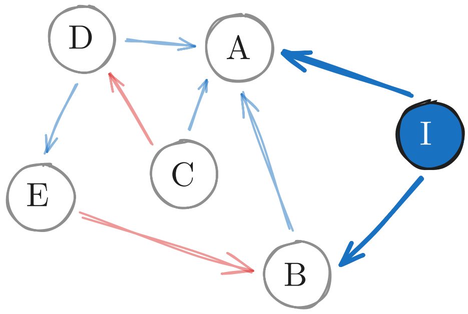
3.5 Modelling interventions

We now outline our approach to modelling interventions in the asymmetric belief system model. In this study we consider interventions which affect the *state* of a belief system. Notably, this excludes interventions intended to influence the structure of a belief system, for instance, by changing the existence of effect size of influence relations between beliefs and attitudes.

Let $\mathcal{M}$ be a belief system model with parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$:



We can consider an intervention as an auxiliary node, I, in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs and attitudes:



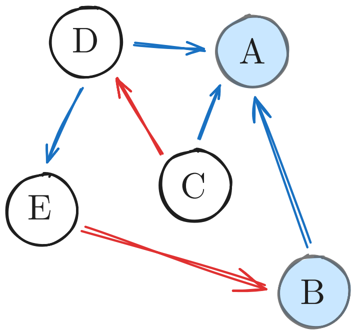
The intervention node I exerts influence on the belief system nodes A and B. Let us consider the effective baseline activation at node A and time $t+1$, as defined in Equation 3.4, given a previous configuration $\boldsymbol{s}^t$:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = h_A + \sum_{\text{node } j} A_{jA} J_{jA} s_j^t \tag{3.10}$$

Since the state of I is fixed (in this case, at $+1$), the contribution of the intervention node to A's effective activation baseline is constant. We can then re-write Equation 3.10, interpreting the intervention effect as an adjustment to the baseline activation h_A:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} A_{jA} J_{jA} s_j^t \tag{3.11}$$

It follows then that we can model interventions more simply as an adjustment to the baseline activation of certain beliefs and attitudes:



Let us define this more formally. For a belief system model $\mathcal{M}$ with $N \in \mathbb{N}$ nodes, and parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$, an *intervention*, is the function $\varphi_{\mathcal{M}} : \delta_h \mapsto \mathcal{M}'$, where $\delta_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}' \rangle$, where

$$\boldsymbol{h}' = \boldsymbol{h} + \delta_h \tag{3.12}$$

TODO: Include discussion on the choice of δ_h, with reference to a figure showing how the activation probability depends on h_i^{eff}.

Chapter 4

Methods

4.1 TODO

- Describe how the derivatives of the objective function change in the symmetric model.

4.2 Counterfactual interventions

Individuals have different belief systems. Influence relations between beliefs and attitudes reflect heterogeneity in individual experiences and perception. Often these relations can themselves be considered beliefs — whether or not an individual who supports climate change mitigation also supports a specific policy depends in-part on their belief regarding the climate impacts of the policy target. An individual's baseline activations, in turn, reflect a culmination of factors which determine their tendency toward certain beliefs or attitudes.

Our approach to parameter estimation, however, fundamentally assumes that the inferred belief system is shared by all individuals in the dataset from which the model is estimated. We note that this is not strictly a limitation of the parameter estimation method, nor of the model itself. Rather this assumption is imposed by data limitations. Fitting individual models requires substantial data from each individual, which is not available in the climate attitudes dataset. Moreover, even with substantial observational data, we are unlikely to observe all state transitions necessary to understand the *potential* dynamics of an individual system, owing to the relatively slow-moving nature of beliefs and attitudes.

For any given individual, the inferred models may thus not accurately reflect the stability of belief system states. A configuration which is stable in a given individual's own belief system may be considered unstable in the shared model. When simulating interventions, we take the most recent measurement for each individual as their initial state. However, due to this perceived instability, we expect to observe natural dynamics away from this initial state, even in null-intervention scenarios.

We thus consider intervention effects as relative to the null-intervention counterfactual baseline rather than relative to the initial state. For each intervention we run both intervention and null ($\delta_h = 0$) scenarios with identical random number generation settings, and define the *intervention effect* as the difference between the observed outcomes.

4.3 Parameter estimation

For the purposes of the experiments in the following sections, we perform parameter estimation to calibrate the belief system model described in Chapter 3.2 to the climate attitudes dataset (Chapter 8). We first outline the context of the parameter estimation problem, and then present our approach. We omit derivation details in this section; however, the curious reader can find these in Appendix A (**TODO**).

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs or attitudes with pre-defined adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N\times N}$, for which the parameters (i.e., the baseline activations and interaction effects) are unknown. Suppose that for each individual $m \in [1, M]$ we have observed a series of measurements, reflecting that individual's belief system state at each of $t \in [1, T]$ uniformly spaced timesteps:

$$\left\{\boldsymbol{x}^t_{(m)}\right\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{x}^t_{(m)} \in \mathbb{R}^N \tag{4.1}$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D, which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

While maximum likelihood estimation (MLE) is often used to infer parameters for similar models (Lee et al., 2015; Nguyen et al., 2017), we should be cautious, for two reasons, of applying this method naïvely in the present study.

Firstly, the model defined in Equation 3.8 of Chapter 3.2 assumes binary spin states $s \in \{-1, +1\}$. The climate attitudes dataset does not satisfy this assumption and must be binarised for MLE to be applicable. However, we have no guarantee that the parameterisation inferred for any specific binarisation is a reasonable explanation for the non-binarised dataset, nor for any other possible binarisation in the case where a probabilistic binarisation scheme is used (such as the one described in Chapter 8.7).

Secondly, maximum likelihood estimation is prone to overfitting when the number of model parameters is similar to the number of observations (Epskamp et al., 2018). While we consider only a small number of beliefs and attitudes in this study ($N \in \{7, 8\}$), the number of parameters p grows rapidly, with $p \in O(N^2)$ for both the symmetric and asymmetric model variants. In combination with sampling error, we expect the number of parameters to negatively impact the robustness of the inferred parameters — given an alternative dataset of the same size, we would expect significant variation in parameters.

We mitigate these respective issues in our parameter estimation scheme by (i) marginalising over the set of possible binarisations of D, and (ii) applying regularisation based on parameter magnitudes. We choose the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which satisfies the optimisation problem:

$$\boldsymbol{\theta}^* = \underset{\hat{\boldsymbol{\theta}} \in \mathbb{R}^p}{\operatorname{argmax}} f\left(D; \hat{\boldsymbol{\theta}}\right), \quad \text{where} \quad f\left(D; \hat{\boldsymbol{\theta}}\right) := \mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) - \lambda \sum_{\theta \in \hat{\boldsymbol{\theta}}} \sqrt{\theta^2 + \varepsilon} \tag{4.2}$$

The first term in the objective function f, $\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right)$, denotes the *expected* log-likelihood given a parameterisation $\hat{\boldsymbol{\theta}}$, over possible binarisations of D:

$$\mathcal{L}_D\left(\hat{\boldsymbol{\theta}}\right) = \mathbb{E}\left[L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \mid \mathcal{D} = D\right] = \sum_{D_B} P(\mathrm{Bin}(\mathcal{D}) = D_B \mid \mathcal{D} = D) \cdot L_{D_B}\left(\hat{\boldsymbol{\theta}}\right) \tag{4.3}$$

where $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ is the log-likelihood of a specific binarisation D_B. We will defer explicitly defining $L_{D_B}\left(\hat{\boldsymbol{\theta}}\right)$ for now, but will return to this point shortly in Equation 4.4.

TODO: Intuition for the effect of using the expectation.

- Inferred model reflects ambiguity in states close to zero.
- The likelihood contribution of each data value is the expectation over possible spin states.
- Values close to zero contribute a comparable weighting from each binary state.
- When the likelihood given one binary state would be significantly higher than for the other, this is downweighted to account for the probability that the value does not obtain this state.
- Therefore the expectation prevents overconfident optimisation toward any specific binarisation — the inferred model is required to explain the binarised dataset *in expectation*.

- In other words, if a data value may be binarised to -1 or $+1$ with equal probability, then both cases should be reasonably explained by the parameterisation.

The second term is a smooth variant of L1 regularisation (Tibshirani, 1996). This has the effect of penalising nonzero parameters which do not meaningfully contribute to the expected likelihood. Parameters instead must reflect relationships which are sufficiently prevalent in the data, reducing the robustness issues resulting from sampling error and a large number of parameters. The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. As $\varepsilon \to 0^+$ the second term converges to the standard formulation of L1 regularisation. Unlike the standard formulation, when $\varepsilon > 0$ the first-derivative of this term exists, so is amenible to gradient-based optimisation. We discuss the choice of values for λ and ε in Chapter 4.3.1.

Let $D_B \sim \mathrm{Bin}(D)$ be a possible binarisation of D, and $\boldsymbol{\theta}$ a parameterisation for the (asymmetric or symmetric) belief system model, which is decomposable into the model parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$. The log-likelihood of D_B given $\boldsymbol{\theta}$ is:

$$\begin{aligned} L_{D_B}(\boldsymbol{\theta}) &= \log P(D_B \mid \boldsymbol{\theta}) \\ &= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \log s_{i,(m)}^{t+1} \cdot h_i^{\mathrm{eff}}\left(\boldsymbol{s}_{(m)}^{t}\right) - \log\left(2 \cosh h_i^{\mathrm{eff}}\left(\boldsymbol{s}_{(m)}^{t}\right)\right) \end{aligned} \tag{4.4}$$

The derivation of Equation 4.4 is analogous to that of the non-equilibrium Ising model log-likelihood (Nguyen et al., 2017). Combining Equation 4.4 and Equation 4.3, we obtain an explicit expression for the expected likelihood:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \left(\sum_{m=1}^{M} \sum_{t=1}^{T-1} P\left(\boldsymbol{S}_{(m)}^{t} = \boldsymbol{s} \mid D\right) \cdot \mathbb{E}\left[\boldsymbol{S}_{(m)}^{t+1} \mid D\right]^T h^{\mathrm{eff}}(\boldsymbol{s}) \right) - Z(D; \boldsymbol{\theta}) \tag{4.5}$$

where $\mathbb{E}\left[\boldsymbol{S}_{(m)}^{t+1} \mid D\right]$ is the expected binary configuration for the observation $\boldsymbol{x}_{(m)}^{t+1}$, the vector $h^{\mathrm{eff}}(\boldsymbol{s})$ contains the effective baseline activation for each spin given the previous state $\boldsymbol{s}$, and Z is the expected log partition function, summed across observations:

$$Z(D; \boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t < T}} P\left(\boldsymbol{S}_{(m)}^{t} = \boldsymbol{s} \mid D\right) \sum_{i=1}^{N} \log\left(2 \cosh h_i^{\mathrm{eff}}(\boldsymbol{s})\right) \tag{4.6}$$

Per the optimisation problem defined in Equation 4.2, we choose the parameterisation $\boldsymbol{\theta}^*$ which maximises the regularised expected log-likelihood, which occurs when $\frac{\partial f}{\partial \theta} = 0$ for every parameter $\theta \in \boldsymbol{\theta}^*$. The partial derivative with respect to a given baseline activation parameter h_i, for $i \in [1, N]$, is given by:

$$
\begin{aligned}
\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m\le M\\ t<T}} & P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\
& - \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}}
\end{aligned} \tag{4.7}
$$

Note that the expressions for $\mathcal{L}_D(\boldsymbol{\theta})$ and $\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta})$ are applicable to both the asymmetric and symmetric belief system models. This property does not hold for the partial derivatives with respect to interaction effects. In the asymmetric model, for $i, j \in [1, N]$, this is derived analogously to Equation 4.7 as:

$$
\begin{aligned}
\frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m\le M\\ t<T}} & P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) A_{ji} s_j \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\
& - \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}}
\end{aligned} \tag{4.8}
$$

In the symmetric belief system model, since we require that $\boldsymbol{J} = \boldsymbol{J}^T$, we estimate a single interaction parameter $J_{ji} = J_{ij} = \beta_{\{i,j\}}$ for each pair of spins S_i, S_j. Each pair thus contributes twice to the partial derivative, once for each direction:

$$
\begin{aligned}
\frac{\partial}{\partial \beta_{\{i,j\}}} f(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m\le M\\ t<T}} & P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{(i',j')\in\{i,j\}} A_{j'i'} s_{j'} \Big[\mathbb{E}\Big[S^{t+1}_{i',(m)} \mid D\Big] - \tanh\big(h_{i'}^{\text{eff}}(\boldsymbol{s})\big)\Big] \\
& - \frac{\lambda \beta_{\{i,j\}}}{\sqrt{\beta_{\{i,j\}}^2 + \varepsilon}}
\end{aligned} \tag{4.9}
$$

4.3.1 Hyperparameters

The optimisation problem in Equation 4.2 has two hyperparameters which must be specified prior to parameter estimation: the regularisation strength $\lambda \in \mathbb{R}^+$, and the smoothing parameter $\varepsilon \in \mathbb{R}^+$.

We use the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008) to select λ, as recommended by Epskamp et al. (2018):

$$
\text{EBIC}(\boldsymbol{\theta}^*) = k \cdot [\ln(M(T-1)) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\boldsymbol{\theta}^*) \tag{4.10}
$$

The variable k denotes the number of nonzero parameters in the optimised model, where a parameter θ is considered nonzero if $|\theta| > 10^{-2}$. We take $\gamma = 0.25$, which has been shown to work well in general for network inference in the inverse Ising problem (Barber & Drton, 2015).

The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observatons are comparable (Foygel & Drton, 2010). We select λ which minimises the EBIC for the symmetric and asymmetric models (independently), the results of which are displayed in Figure 4.1.

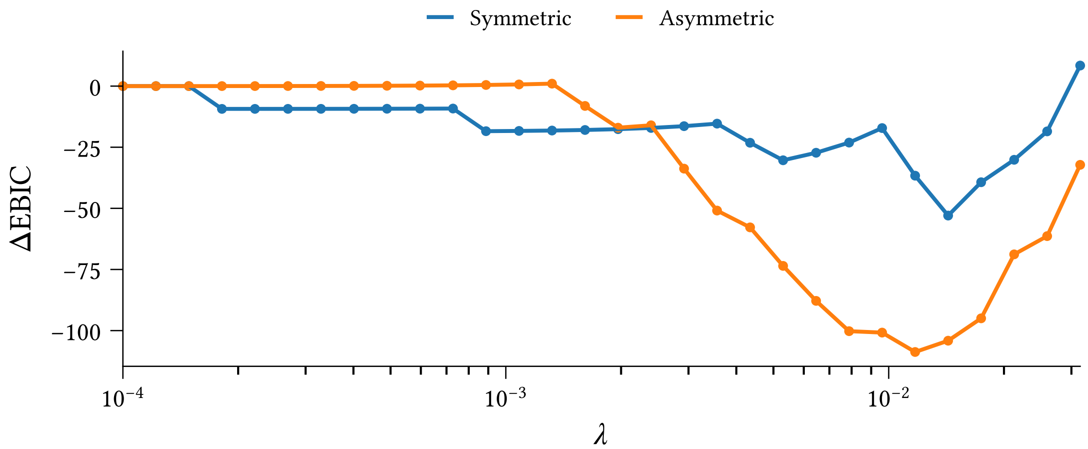


Figure 4.1: Effect of regularisation strength λ on Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate attitudes dataset using Equation 4.2. The vertical axis measures the difference in EBIC compared to the no-regularisation case ($\lambda = 0$).

For the purposes of this study we consider parameters with magnitude less than 10^{-2} as 'effectively zero'. We choose $\varepsilon = 10^{-8}$ such that $\sqrt{\varepsilon}$ is significantly smaller than this threshold, and $\sqrt{\theta^2 + \varepsilon}$ remains a good approximation to the standard formulation.

4.3.2 Implementation details

We use the Scipy 1.17.1 implementation (Virtanen et al., 2020) of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136) to solve the parameter estimation problem, using the jacobian comprising the parameter-specific partial derivatives. We take take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess.

To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps this amounts to a scale factor of $\frac{1}{M(T-1)}$.

Chapter 5

Existence and impact of asymmetry in belief systems

5.1 Plan

First results section, for experiments on the effect of asymmetry

- **RF1:** Show asymmetric model fit
- **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
- **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).

5.2 Fitting model

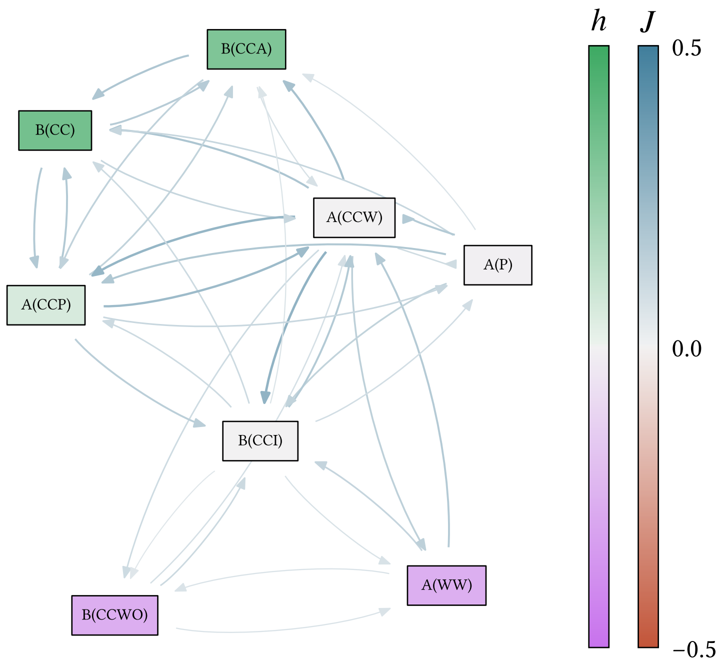


Figure 5.1: TODO

5.3 RQ2.1: Asymmetry in estimated models

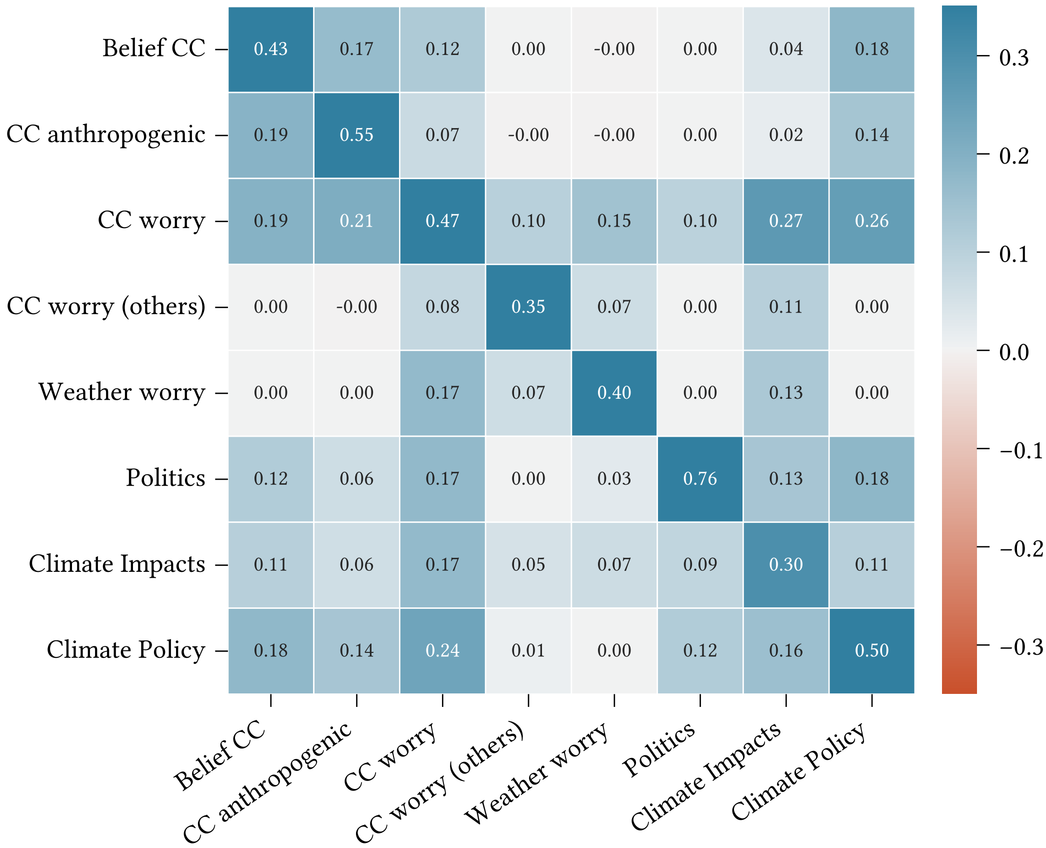


Figure 5.2: Climate belief system interaction effect matrix.

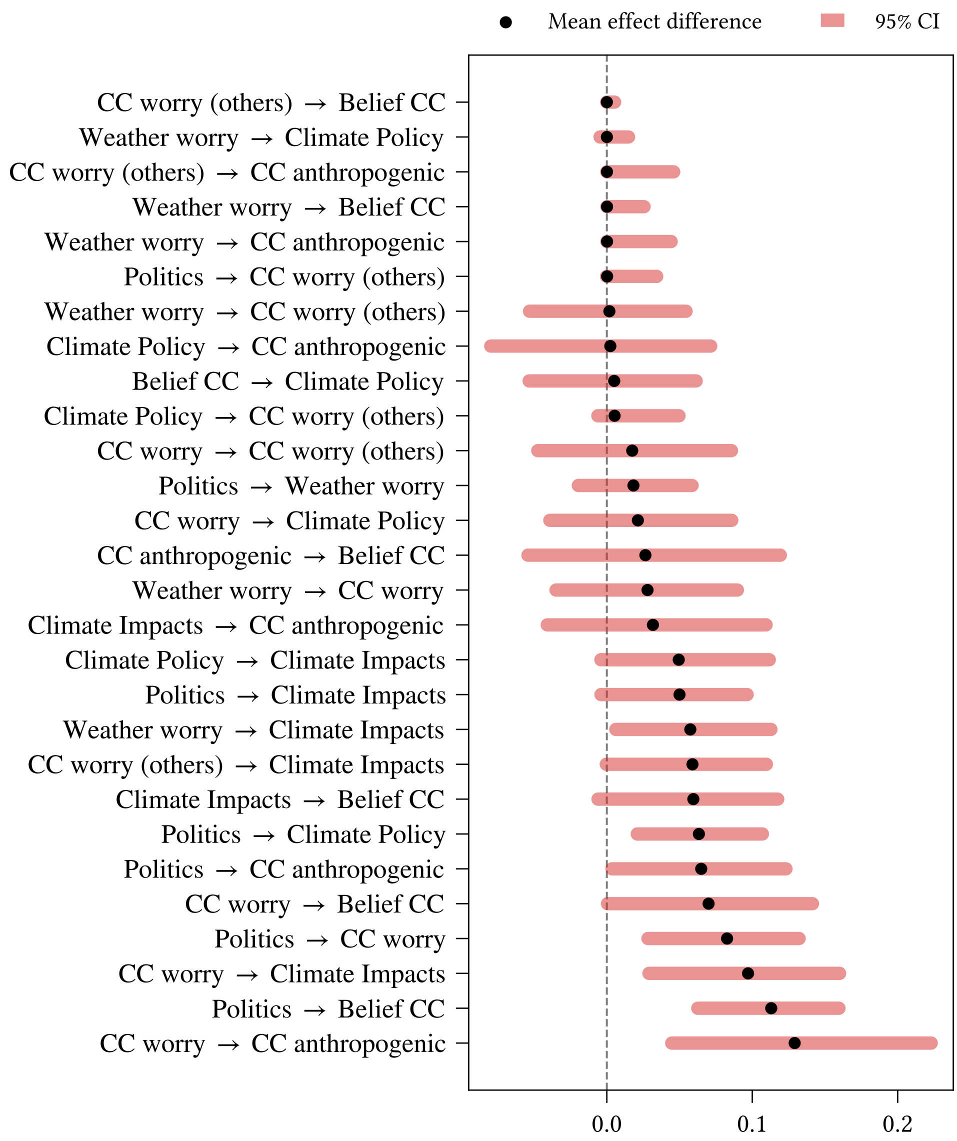


Figure 5.3: TODO

5.4 RQ3.1: Intervention strategy and effectiveness under sym/asym assumptions

Fit models using bootstrapping. Consider both effects of intervention (how does intervening at S affect other spins) and strategies (how does intervening at different spins affect this *one*). Repeat for varying levels/strengths of intervention.

For strategy, estimate expected ranking of interventions at both collective and individual levels. For the former, measure collective effect, then rank interventions, then take mean across repeats. For the latter, measure expected individual effect, rank interventions, then take mean across individuals.

5.4.1 Ranking interventions

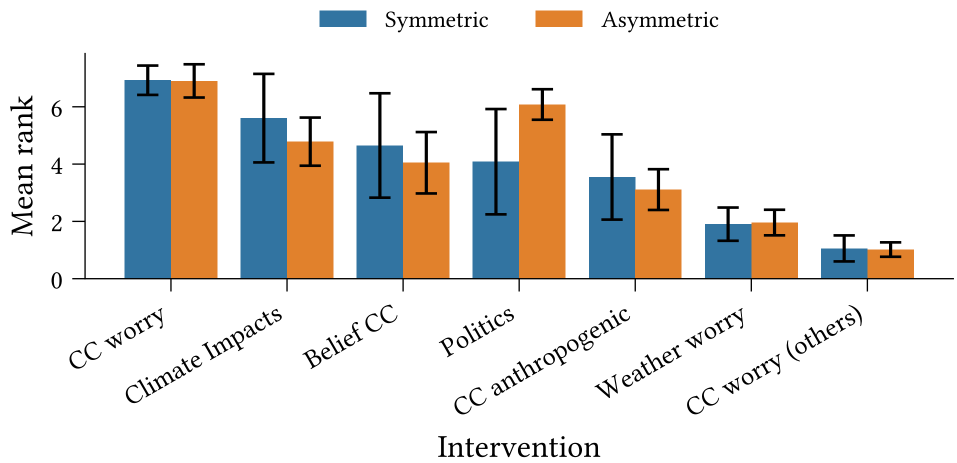


Figure 5.4: Expected ranking for weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

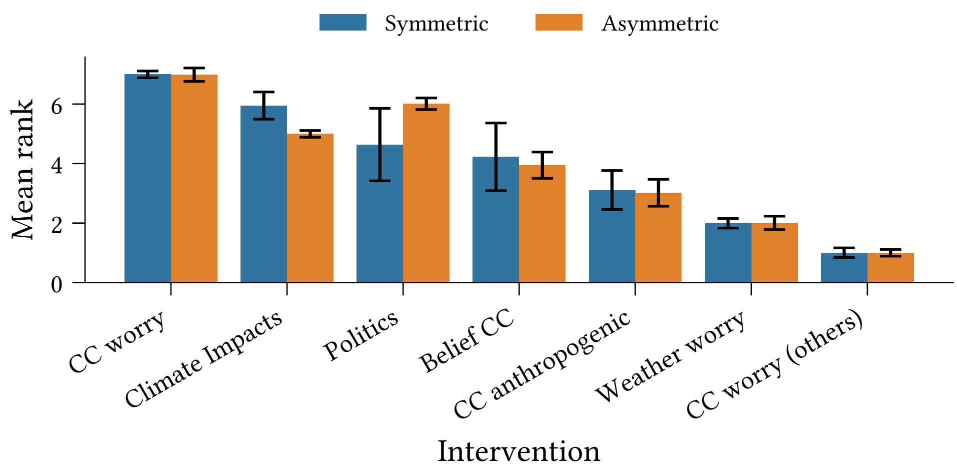


Figure 5.5: Expected ranking for medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

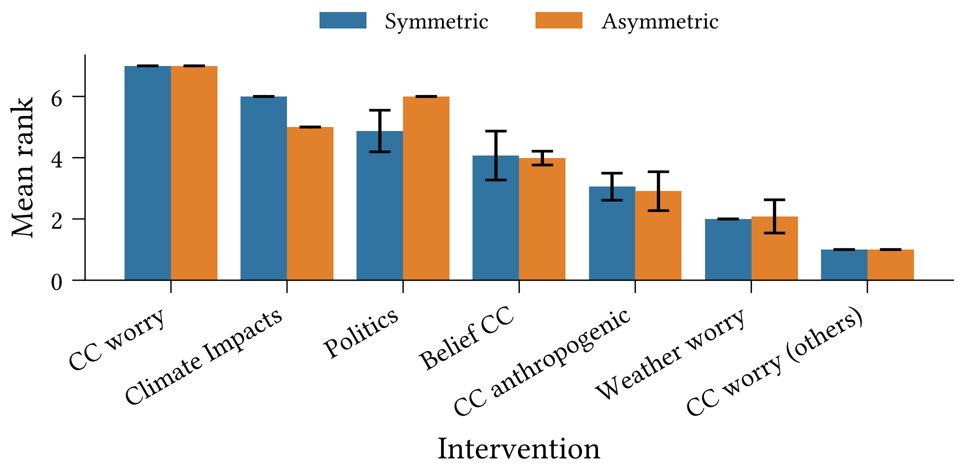


Figure 5.6: Expected ranking for perfect interventions ($\mathrm{do}(S_i = +1)$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

5.4.2 Intervention effect

Measures the average difference in state between intervention and no-intervention scenarios

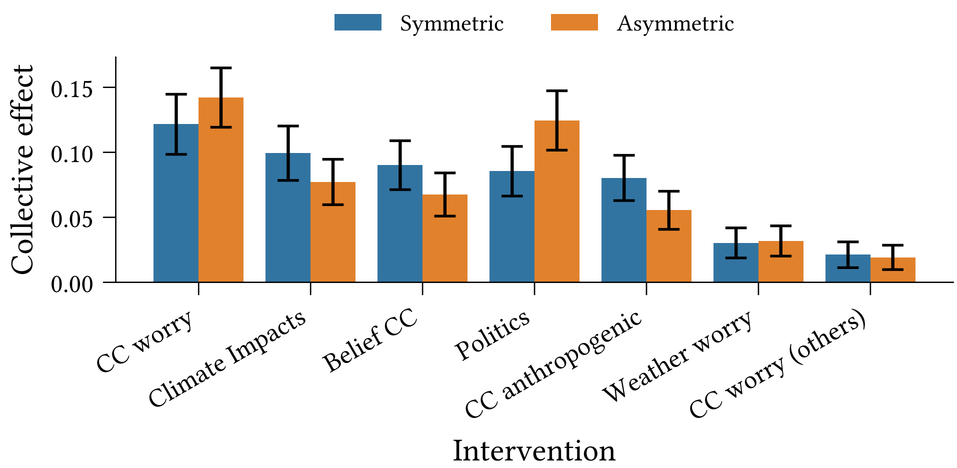


Figure 5.7: Expected effect of weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

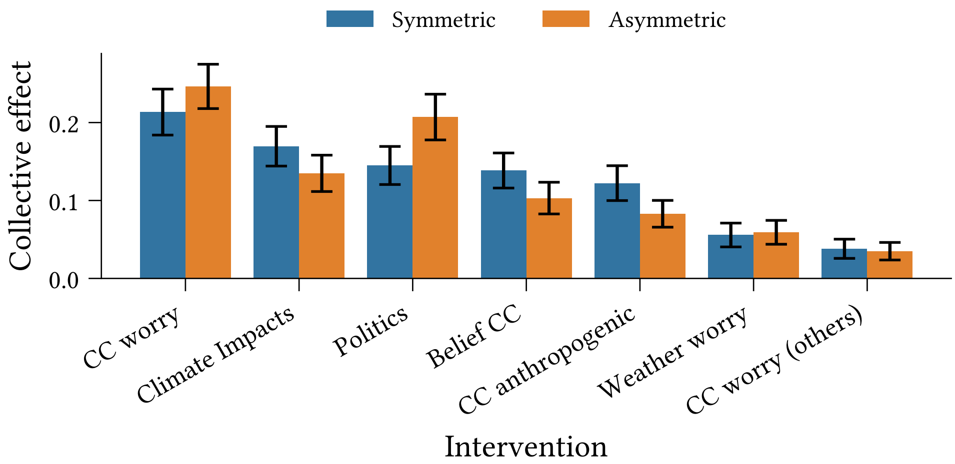


Figure 5.8: Expected effect of medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

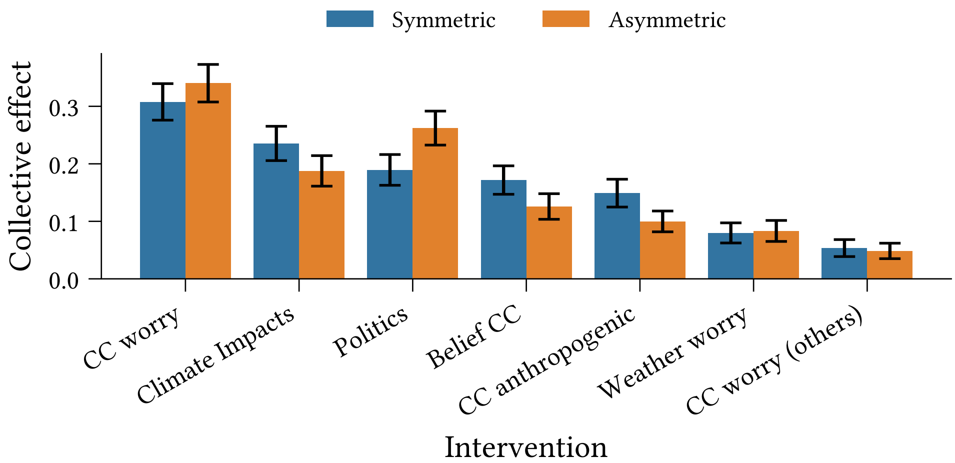


Figure 5.9: Expected effect of perfect interventions ($\mathrm{do}(S_i = +1)$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

Chapter 6

Heterogeneous belief systems and intervention effects

6.1 Plan

Second results section, on individual heterogeneity in belief system structure, and intervention dynamics.

- **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.
- **RQ3.2:**
  - Distribution of intervention effectiveness by individual.
  - How does intervention ranking vary across individuals?
  - Characterising how different initial states affect intervention success.
  - Looking at how other theory-driven features affect success, e.g.:
    - How receptive is the individual to the intervention?

      The effective baseline $(h_i + \sum_j J_{ji} s_j)$ determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

      We could look at how this changes with different intervention strengths (it follows a logistic curve).

    - How consistent is the individual's belief state, as measured by the total system energy?

– How ‘entrenched’ is the target attitude?

Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

6.2 RQ2.2: differences between belief systems for different (types of) individuals

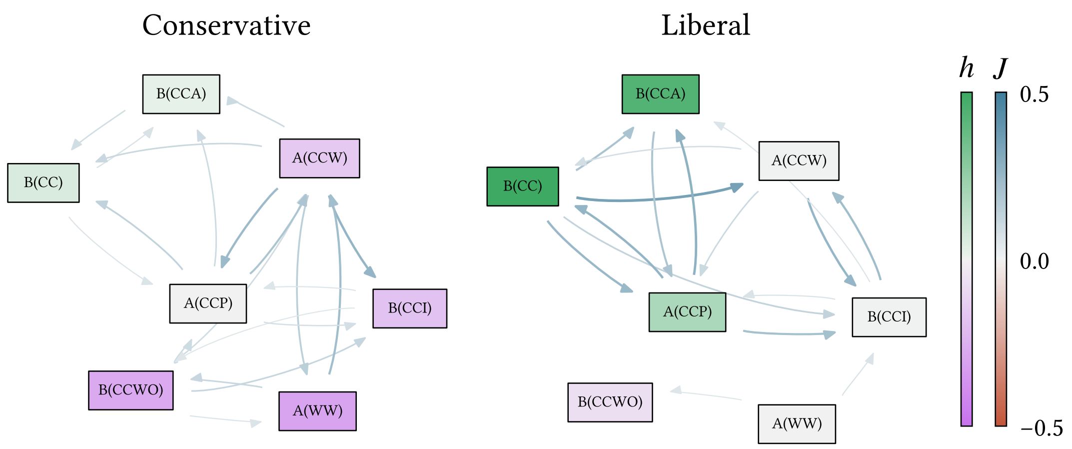


Figure 6.1: Prefixes: A (attitude), B (belief); node labels: CC (climate change), CCA (climate change anthropogenic), CCW (climate change worry), CCWO (climate change worry others), CCI (climate change impacts), CCP (climate change policies), WW (weather worry).

6.3 RQ3.2: Intervention strategy and effectiveness across individuals

Examine distributions of expected outcome effects (per individual) for different interventions. Characterise the individuals found at different parts of the distribution (personas, average magnetisation).

6.3.1 Intervention ranking

For each individual, compute ranking over the expected effects of each possible intervention.

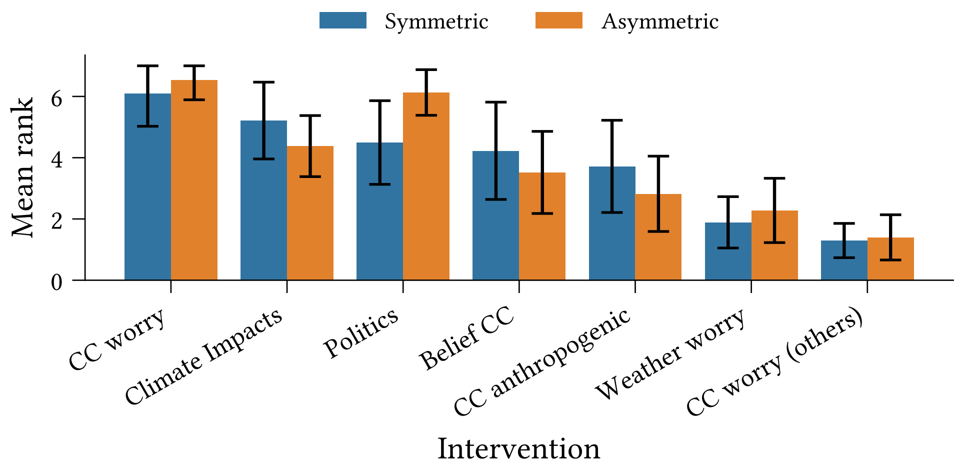


Figure 6.2: Ranked intervention effects per-individual, weak, targeting climate policy

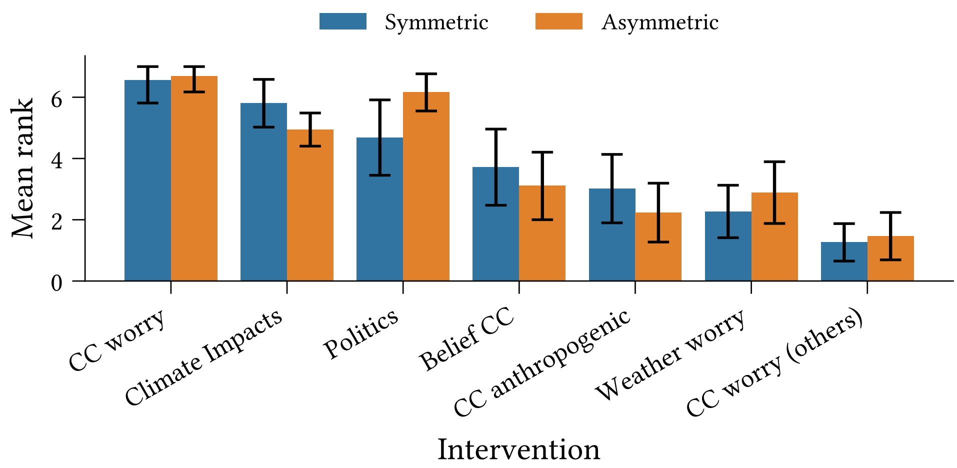


Figure 6.3: Ranked intervention effects per-individual, perfect, targeting climate policy

Chapter 7

Discussion

7.1 Plan

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
  - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
  - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
  - More generally, effect sizes may vary depending on specific spin states.

Chapter 8

Climate beliefs dataset

8.1 Plan

- Introduce the (insert name here) dataset, give context, survey details
- Validation, cleaning, transformations
- Question selection, indexes
- Binarisation

8.2 Longitudinal climate attitudes survey

Introduce the survey:

- Motivate it: This is the main resource used to construct our dataset.
- Description:
  - Content: What sorts of questions? (Examples)
  - Waves: how many, when were they run? (Participation event plot)
  - Participation: counts, variability. (Statistics: how many people answered all waves? How many singletons?)
  - Question variability. (...?)
- Overlap with notable events:
  - COVID-19 outbreaks
  - US electoral cycle
  - Major weather events

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and

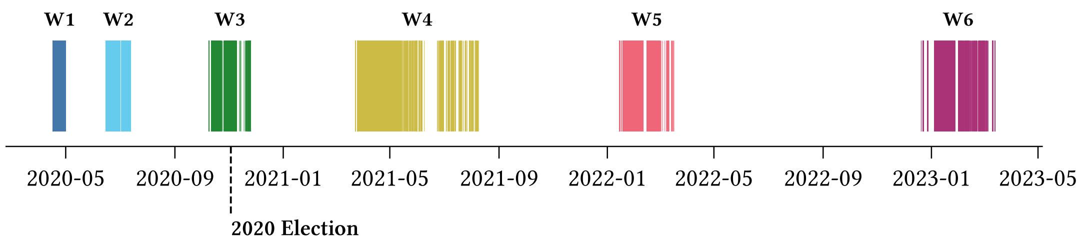


Figure 8.1: Longitudinal survey response dates per-wave

financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies.

- Significance of longitudinal data — most studies use cross-sectional data.

Survey content; nature of the questions asked

- Context
- Category (belief, attitude, demographic, etc.)
- Response formats

The survey content varies between waves, and between participants.

Participation

Waves The survey waves occur with irregular spacing and duration, as illustrated in Figure 8.1. Some consecutive pairs of waves are much closer than others. For instance, the joined interval spanned by the first three waves has a longer duration than the interval between Waves 5 and 6, and the duration spanned by Wave 4 exceeds that of the joined interval comprising Waves 1 and 2.

The 2020 US Presidential Election date occurred during Wave 3. This wave was modified for responses started after the election date; future-focused survey questions regarding individuals' voting intentions, beliefs, and attitudes concerning the election were removed.

- Context of the survey:
  - Time-span
  - Wave timing/duration
  - Overlap with COVID-19 outbreaks and US electoral cycle, major weather events
- Codebook covers first five waves
- Nature of the questions asked:
- Variability in questions (between waves, participants)

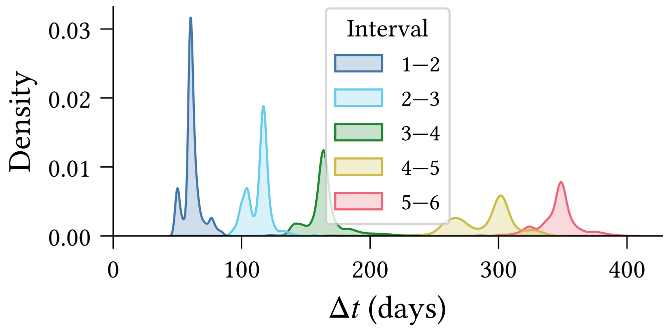


Figure 8.2: Between-response time distribution

8.3 Question selection

- Motivation (identify measures of distinct beliefs and attitudes regarding climate change – impacts, risks, policies, etc.)
- Criteria for selecting questions
- Exclude sixth wave due to no codebook

8.4 Validation

- Only validate selected questions
- Validation components:
  - Schema (Pandera)
  - Null responses iff null expected
  - Valid survey responses
  - Miscellaneous inconsistencies
    - Born in US in some waves, not in others
    - Changing responses to extreme weather experience in last 10 years
- Results:
  - Schema good
  - Various issues with Null responses:
    - Some due to survey logic errors
    - Some not explained

8.5 Normalisation, cleaning, and transformations

- Variable names (replace '.' with double underscore)
- Filter out participants with:
  - Null `participant_id`

  - Survey errors (from validation above)
  - Failed survey validation checks (if any check is failed, participant is removed entirely)
- Response type/value conversions
  - Non-float numeric variables to Int64
  - Re-map $1.. = k$ variables to $(0.. = k - 1)$
  - Replace empty string entries with `Null`
  - Multiple choice questions: comma-separated string of integers to list of integers/ enums
  - Replace categorical variable integer responses with enums (for human-readability and to ensure value consistency across waves)
  - `cc2`: Re-order such that human-causes are 'high values' and natural-causes/no climate change are 'low values'
- Coalescing treatment columns:
  - Responses stored in one column
  - Separate index column specifies treatment group membership
  - Optional third column for treatment-specific numeric parameters (e.g., policy cost)
- Constructed columns:
  - Merge `pol_party` and `pol_lean` to create five-point `pol_affiliation` scale. Note that this does not capture individuals who 'lean *right* toward Democrat' for example.

8.6 Data Validation

The complexity of the climate attitudes survey (**reference section discussing this**) necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.

2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.

3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

In the first instance, type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions according to the question type (see Table 8.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type[1]. Type validation for a multiple-choice question thus requires checking that the response column comprises `list`'s whose elements are all from the set of values defined by the `Enum`.

Response-value validation then ensures that all non-null responses are valid according to the set of allowable values described in the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle with schema variation between waves.

We implement both type-level and response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). The Pandera API allows for straightforward schema specification using Python type annotations (for type-level validation) in combination with value constraints (for response-level validation). We implement a separate validation check for questions whose response schema varies between survey waves, using a Pandera 'dataframe check' to stratify validation by wave.

Null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W can be null for one of four reasons:

Response schema	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 8.1: **TODO**

[1]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\text{Urban}, \text{Suburban}, \text{Rural}\}$.

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4}) \tag{8.1}$$

We split this up into: If not in wave, then R is null.

$$(\text{N1} \lor \text{N2}) \implies \text{null}(R) \tag{8.2}$$

If conditions not met, then R is null.

$$(\text{N3} \lor \text{N4}) \implies \text{null}(R) \tag{8.3}$$

For unconditional columns, if shown, then not null:

$$\neg(\text{N1} \lor \text{N2}) \implies \neg\,\text{null}(R) \tag{8.4}$$

For conditional columns, if shown, then not null:

$$\neg(\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4}) \implies \neg\,\text{null}(R) \tag{8.5}$$

For a conditional $A \implies B$, let $\boldsymbol{R}$ be the subset of responses for which the contradiction $\neg(A \implies B) \equiv (A \land \neg B)$ is true. Then $A \implies B$ is satisfied if, and only if, $\boldsymbol{R}$ is empty **check this — is it iff, if, or only if?**. If $\boldsymbol{R}$ is non-empty, then the contained responses are counter-examples to the conditional.

- Single-response ordinal:

In the first case, type-level validation ensures that the data schema matches expectations, independently of the response values. In the first case, type-level validation ensures that the schema observed in the dataset matches the schema Type-level validation Our validation pipeline comprises three stages:

The first two stages are handled by Pandera, and the third is manually implemented.

- What does type-level validation consist in?
  - Simple types
  - Enums
  - Lists of enums
- What does response value validation consist in?
  - Checking that observed (non-null) values are within the set of allowable values
- What does null-value validation consist in?
  - ...

The data schema defines everything that is necessary. Maybe we can discuss this separately as an implementation step.

8.7 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., '3' on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 8.3 illustrates this process for a negative value of x and given choice of σ.

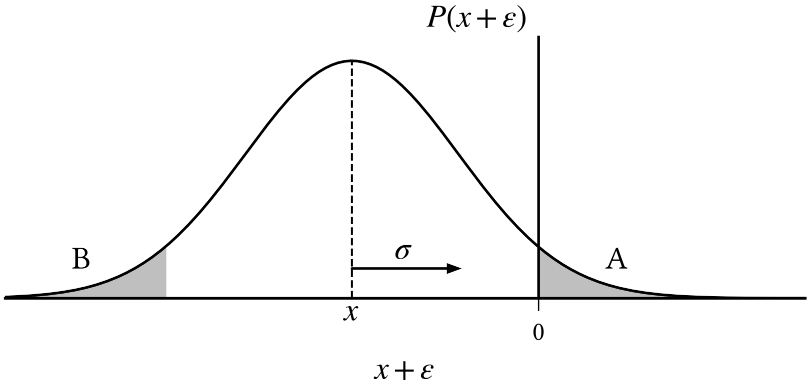


Figure 8.3: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \mathrm{A} = \mathrm{B} = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \tag{8.6}$$

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.2$ such that a ‘weakly oppose’ response to a 7-point Likert scale[2] (value 1/3) is mapped to $+1$ with probability 0.05.

[2]The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

Chapter 9

Related work

Considering moving this section to later in the thesis.

- Early work on modelling belief systems, triadic consistency
- Approaches:
  - partial correlation networks,
  - Bayesian networks,
  - Causal attitude network, Ising-style models
- Beliefs as edges vs. beliefs as nodes
- Social influence: hierarchical Ising models, network of belief model
- Within-person vs between-person correlations; complications around inferring individual belief systems.

Chapter 10

Conclusions and future work

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Chapter 11

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: "The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society". You don't need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission "NDA" is written under "Availability" and "License" for the concerned data items.

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Appendix A

Parameter estimation

Log-likelihood

Equation 4.4

Expected log-likelihood

Equation 4.5

Partial derivatives of expected log-likelihood

Dataset

Binarisation probability

Equation 8.6